

TechMart Electronics

Technical Support Knowledge Base

Covering the TechMart Connect Platform, TechMart Home Hub Devices, and the TechMart Developer API

Sections: Login Issues • Password Reset • Installation Guide • Troubleshooting
• API Errors • System Requirements

Internal Reference Document — Version 3.2 — For Retrieval-Augmented Generation (RAG) Knowledge Base Use

This document is a fictional dataset created for educational and software-development testing purposes only. TechMart Electronics, TechMart Connect, and TechMart Home Hub are fictional product names.

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Note: this knowledge base is organized for chunked ingestion into a vector database. Each numbered subsection and appendix entry above is written to stand largely on its own, making it suitable as an independent retrieval unit for a Retrieval-Augmented Generation pipeline.

Section 1: Login Issues

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This section describes the most common login problems reported by TechMart Connect users across the web portal, the desktop application, and the mobile application, along with the diagnostic steps and resolutions used by the TechMart support team. TechMart Connect is the unified account system used to manage TechMart Home Hub devices, subscriptions, billing, and the TechMart Developer API. Because login sits at the front of nearly every support interaction, agents should treat this section as the first reference point before escalating a ticket.

1.1 Overview of the Authentication System

TechMart Connect uses a token-based authentication system built on OAuth 2.0 with support for both password-based sign-in and single sign-on (SSO) through Google, Apple, and Microsoft accounts. When a customer signs in successfully, the platform issues a short-lived access token valid for 60 minutes and a refresh token valid for 30 days. The refresh token is stored securely on the client device and is used to silently renew the access token without requiring the customer to re-enter their credentials. Login failures typically fall into one of five categories: invalid credentials, account lockout, expired sessions, SSO provider errors, or multi-factor authentication (MFA) failures.

Support agents should always confirm which platform the customer is using (web, iOS, Android, or Windows/macOS desktop client) because the underlying error codes and remediation steps differ slightly between platforms. The web portal reports errors as on-screen banners with a six-digit error code beginning with the prefix AUTH, while the mobile applications surface the same errors as toast notifications with a shortened three-digit code.

1.2 Invalid Credentials

The most frequent login issue is an 'invalid email or password' message, returned as error AUTH-100. This occurs when the email address is mistyped, when the password contains a typo, or when the customer is attempting to log in with an old password that was changed on another device. Before assuming the credentials themselves are wrong, agents should verify that Caps Lock is not enabled, that the customer is not including a trailing space when copying and pasting their password, and that the account region matches the region selected in the login form, since TechMart Connect maintains separate authentication clusters for the Americas, Europe, and Asia-Pacific regions.

If the customer confirms their password is correct but the error persists, the next step is to check whether the account was created using an SSO provider. Approximately 38% of support tickets in this category are caused by a customer attempting to use a password login when their account was originally created through 'Sign in with Google' or 'Sign in with Apple.' In these cases, no password exists on the account, and the customer must use the matching SSO button instead.

1.3 Account Lockout (AUTH-110)

After five consecutive failed login attempts within a 15-minute window, TechMart Connect automatically locks the account for security purposes and returns error AUTH-110. The lockout is time-based and resolves automatically after 30 minutes, but customers who need immediate access can request an unlock link, which is emailed to the address on file and remains valid for 20 minutes. Support agents can also manually clear a lockout from the internal Account Administration Console under Account Actions > Clear Login Lockout, though this should only be done after verifying the customer's identity using the standard three-point identity check (full name, account email, and either the last four digits of the payment method on file or the device serial number of a registered Home Hub).

Repeated lockouts on the same account within a short period may indicate a credential-stuffing attempt from a leaked password database rather than genuine customer error. Agents should check the Security Timeline in the Account Administration Console for login attempts originating from unfamiliar IP addresses or countries. If suspicious activity is found, the agent should force a password reset, revoke all active sessions, and advise the customer to enable multi-factor authentication.

1.4 Expired or Invalid Sessions (AUTH-120, AUTH-121)

Customers occasionally report being logged out unexpectedly, particularly on the desktop application. This typically occurs when the refresh token has expired (AUTH-120), the token has been revoked because the customer changed their password on another device (AUTH-121), or the system clock on the customer's device has drifted by more than five minutes, which causes the token validation to fail. Clock drift is especially common on older Windows machines where automatic time synchronization has been disabled, and on virtual machines that have been suspended for extended periods.

Resolution steps for expired sessions

- Confirm the device's date, time, and time zone are set to automatic.
- Ask the customer to fully sign out and sign back in rather than closing and reopening the app.
- Clear the local token cache: on desktop this is located at %APPDATA%/TechMart/Connect/tokens.db on Windows or ~/Library/Application Support/TechMart/Connect/tokens.db on macOS.
- If the issue persists across multiple devices, check the Security Timeline for a forced session revocation triggered by a password change or a 'Sign out of all devices' action.

1.5 Single Sign-On (SSO) Failures

SSO login failures are reported as error AUTH-200 through AUTH-204, each corresponding to a specific identity provider issue. AUTH-200 indicates the identity provider (Google, Apple, or Microsoft) did not return a valid token, usually because the customer denied the permission prompt or closed the provider's login window before completing authentication. AUTH-201 indicates the email address returned by the identity provider does not match any existing TechMart Connect account and no new account was created because the customer had previously opted out of automatic account creation. AUTH-202 is returned when a customer's corporate Microsoft account has conditional access policies that block third-party application logins; in this scenario, the customer's IT administrator must whitelist TechMart Connect in their organization's Entra ID application list.

AUTH-203 and AUTH-204 relate to Apple's private email relay feature. When a customer signs in with Apple and chooses to hide their real email address, Apple provides a relay address ending in @privaterelay.appleid.com. If the customer later disables the relay from their Apple ID settings, TechMart Connect can no longer deliver account notifications, and the login itself may fail with AUTH-204 if Apple's relay service is temporarily unreachable. Agents should advise customers to add support@techmart-electronics.example to their Apple ID's allowed senders list to prevent delivery failures.

1.6 Multi-Factor Authentication (MFA) Failures

Customers who have enabled MFA may be unable to complete login if their time-based one-time password (TOTP) codes are consistently rejected (AUTH-300). This is almost always caused by clock drift on the authenticator device or app, since TOTP codes are only valid within a 30-second window and TechMart Connect allows a tolerance of one interval before and after the current window. Ask the customer to check that their authenticator app has automatic time sync enabled. If the customer has lost access to their authenticator device entirely, agents can issue one of the ten single-use backup codes generated when MFA was first enabled, or, if those have also been exhausted, initiate the Identity Recovery process described in Section 2.5.

Common MFA error codes

Code	Meaning	Typical Cause
AUTH-300	TOTP code rejected	Clock drift on authenticator device
AUTH-301	Backup codes exhausted	All 10 single-use codes already redeemed
AUTH-302	SMS code not received	Carrier delay or incorrect phone number on file
AUTH-303	MFA device unenrolled	Customer reset their phone without transferring MFA

1.7 Browser-Specific Login Issues

The TechMart Connect web portal is officially supported on the latest two major versions of Chrome, Firefox, Edge, and Safari. Login failures that only occur in a browser are frequently related to blocked third-party cookies, aggressive tracking-protection extensions, or corporate content filters that intercept the OAuth redirect. Safari's Intelligent Tracking Prevention (ITP) has historically caused issues with the SSO redirect flow because it partitions storage per top-level site; customers using Safari 17 or later should be directed to enable 'Allow website to use cookies to keep me signed in' under Safari Settings > Privacy.

- Ask the customer to try an incognito or private browsing window to rule out extensions and cached cookies.
- Confirm that JavaScript is enabled, since the login form performs client-side token exchange.
- Check whether the customer is on a corporate or school network that may block outbound requests to auth.techmart-electronics.example on port 443.
- For persistent Safari ITP issues, recommend using the TechMart Connect desktop or mobile app instead of the browser.

1.8 Regional and Data Residency Login Errors

TechMart Connect stores customer data in one of three regional clusters based on the country selected during signup: Americas (us-east), Europe (eu-central), and Asia-Pacific (ap-southeast). Error AUTH-400 occurs when a customer attempts to log in from a region-locked domain that does not match the cluster their account was created in, for example logging in at eu.techmart-connect.example with an account created on the Americas cluster. The fix is simply to redirect the customer to the correct regional URL, which can be looked up in the Account Administration Console using the customer's email address. Agents should never attempt to manually migrate an account between clusters, as this must be done through the Data Residency Migration process owned by the Platform Engineering team.

1.9 Escalation Criteria for Login Issues

Most login issues are resolved at Tier 1 using the steps above. A login ticket should be escalated to Tier 2 Technical Support when any of the following are true: the customer's account shows evidence of unauthorized access after a password reset, the SSO provider issue persists after the identity provider has confirmed the account is in good standing, the customer is affected by a suspected regional outage (check the internal Status Dashboard first), or the login failure is reproducible on a brand-new test account, indicating a platform-wide bug rather than an account-specific issue.

1.10 Login Rate Limiting and Bot Protection

To protect against automated credential-stuffing attacks, the login endpoint applies an IP-based rate limit independent of the per-account lockout described in Section 1.3. An IP address that submits more than 30 login attempts across any accounts within five minutes is temporarily blocked at the network edge for 15 minutes, returning error AUTH-500. This most commonly affects customers on carrier-grade NAT networks, such as certain mobile carriers or large office buildings, where many unrelated users share a single public IP address. If a customer reports AUTH-500 and identity verification confirms only a small number of legitimate attempts from their own household, agents can submit an IP allowlist exception request through the Trust and Safety queue, valid for 24 hours.

The login form also includes an invisible bot-detection challenge that runs automatically in the background for web portal logins. Customers using aggressive privacy browser extensions that block all third-party scripts, including legitimate security scripts, may see their login silently fail with no error code at all, simply refreshing back to the login form. Advising the customer to temporarily disable strict script-blocking extensions for techmart-electronics.example typically resolves this rare but confusing scenario.

1.11 Legacy Account Migration Login Issues

Customers who created their account before the 2024 platform migration to the current OAuth 2.0-based authentication system may occasionally encounter error AUTH-600, indicating their account record still references the legacy authentication schema and requires a one-time silent migration. This migration happens automatically on the customer's next successful login and is invisible to the customer in the vast majority of cases, but a small number of accounts with unusual characters in their original password hash have failed the automatic migration and require a manual trigger from the Account Administration Console under Account Actions > Force Legacy Migration. This action is safe to perform at any time and does not

affect the customer's existing password or device pairings.

Section 2: Password Reset

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This section covers the self-service password reset flow, the agent-assisted reset flow, password policy requirements, and the identity recovery process for customers who have lost access to both their password and their MFA device.

2.1 Self-Service Password Reset Flow

Customers can reset their own password from the 'Forgot password?' link on the login screen of the web portal, desktop app, or mobile app. The flow requires the customer to enter the email address associated with their account, after which TechMart Connect sends a password reset email containing a unique, single-use link. The link is valid for 20 minutes and expires immediately after use. If the customer does not receive the email within a few minutes, the most common causes are a typo in the email address, the message being filtered into a spam or promotions folder, or the account actually being registered under a different email address than the one the customer expects, often a work email versus a personal email.

Because the reset email is sent from no-reply@techmart-electronics.example, agents should advise customers to check their spam filter and, if using a corporate email account, ask their IT department to allow-list the sending domain. If a customer reports never receiving reset emails across multiple attempts, agents should check the Email Delivery Log in the Account Administration Console for hard bounces, which indicate the mailbox does not exist or is rejecting messages from the TechMart mail servers.

2.2 Password Policy Requirements

TechMart Connect enforces a password policy that requires a minimum of 10 characters, including at least one uppercase letter, one lowercase letter, one number, and one symbol. Passwords are checked against a database of over 900 million previously breached passwords at the time of creation; if a match is found, the reset is rejected with error PWD-010 even if the password otherwise meets the complexity requirements. Passwords cannot match any of the customer's last five previously used passwords (error PWD-011), and passwords cannot contain the customer's email address, first name, or last name as a substring (error PWD-012).

Password error codes

Code	Meaning
PWD-010	Password found in known breach database
PWD-011	Password matches one of the last five used passwords
PWD-012	Password contains personal information
PWD-013	Password does not meet minimum complexity requirements
PWD-014	Reset link expired or already used

2.3 Agent-Assisted Password Reset

When a customer cannot complete the self-service flow, for example because the email account itself has been lost, an agent may initiate an assisted reset from the Account Administration Console under Account Actions > Send Password Reset. This still requires the customer to click a link sent to the email on file; TechMart Connect does not allow agents to set a password directly on behalf of a customer under any circumstance, as this would bypass identity verification. If the customer has also lost access to their email account, the ticket must be routed to the Identity Recovery process described in Section 2.5 rather than processed as a standard password reset.

Before initiating any assisted reset, the agent must complete the standard identity verification: full legal name matching the account, the email address on file, and one additional identifier such as the last four digits of the payment method, a recent order number, or the serial number of a registered TechMart Home Hub device. Agents should never disclose whether a given email address has an associated TechMart account to anyone who cannot pass identity verification, as this is considered a data-privacy violation under company policy.

2.4 Mobile App Password Reset Considerations

On iOS and Android, the password reset link opens in the device's default browser rather than directly inside the TechMart app due to platform restrictions on deep-linking from email clients. After setting a new password, customers must manually reopen the TechMart app and sign in again; the app does not automatically detect the completed reset. A common point of confusion is that customers using the Gmail or Outlook mobile apps sometimes have in-app browsers that do not preserve the session correctly. In these cases, ask the customer to open the reset link in Safari or Chrome directly rather than the email app's built-in browser.

2.5 Identity Recovery for Lost Email and MFA Access

Customers who have lost access to both their registered email address and their MFA device require the full Identity Recovery process, which is intentionally slower than a standard reset to prevent account takeover. The customer must submit a government-issued photo ID, a photo of themselves holding the ID, and at least two of the following: a recent billing statement showing a TechMart charge, the serial number of a registered TechMart Home Hub device, or a screenshot of TechMart order history from the associated retailer account. Identity Recovery requests are reviewed by the Trust and Safety team within three to five business days and cannot be expedited by support agents, regardless of the customer's frustration or urgency; agents should set this expectation clearly and empathetically at the start of the process.

2.6 Bulk Password Reset for Organizational Accounts

TechMart Connect for Business customers who manage multiple Home Hub devices under a single organizational account can trigger a bulk password reset for all linked sub-accounts from the Admin Console under Organization > Security > Force Password Reset. This immediately invalidates all active sessions for every user in the organization and requires each user to set a new password at their next login. This feature should be recommended to business customers who suspect a shared password has been compromised, such as after an employee departure.

2.7 Password Managers and Autofill Issues

Customers using third-party password managers occasionally report that the reset form rejects a newly generated password even though it appears to meet all complexity requirements. This is almost always caused by the password manager generating a password longer than the platform's 64-character maximum, or including a character such as a curly quote produced using a non-standard character encoding. Recommending the customer manually cap the generated password length to under 64 characters resolves the vast majority of these cases. Agents should also confirm the customer's password manager browser extension is not auto-submitting the form prematurely, before the confirmation field has been populated, which can appear as a generic validation failure.

2.8 Temporary Passwords for TechMart for Business Onboarding

When a business administrator creates a new employee account through the Admin Console, the system generates a temporary password that is valid for a single login only and must be changed immediately upon first sign-in. Temporary passwords expire after 72 hours if unused; if a new employee has not logged in within that window, the administrator must generate a new temporary password rather than the employee using the standard self-service reset flow, since no personal email verification exists yet for accounts provisioned this way until the first successful login is completed.

Section 3: Installation Guide

Document ID: TM-TECH-INSTALL-03 | TechMart Electronics Technical Support Knowledge Base | Category: Technical Support

This section walks through installation of the TechMart Connect application across supported platforms, initial Home Hub device setup, firmware provisioning, and license activation for the TechMart Developer API.

3.1 Pre-Installation Checklist

- Confirm the device meets the minimum system requirements listed in Section 6.
- Ensure a stable 2.4GHz or 5GHz Wi-Fi network is available (the Home Hub setup process requires 2.4GHz for the initial pairing step).
- Have the TechMart account email and password ready, or be prepared to create a new account during setup.
- Disable any VPN during the initial pairing process, as some VPN configurations interfere with local network device discovery.
- Locate the setup code printed on the bottom of the Home Hub device or on the included Quick Start card.

3.2 Installing the Windows Desktop Application

Download the installer from downloads.techmart-electronics.example/connect/windows, which provides a signed .exe installer approximately 180MB in size. Administrator privileges are required for installation because the app registers a background service, TechMart Connect Sync Service, which keeps device states synchronized even when the main application window is closed. During installation, Windows SmartScreen may display a warning because the installer is relatively new; customers should click 'More info' and then 'Run anyway' to proceed, and can verify the publisher is listed as 'TechMart Electronics, Inc.' before doing so.

After installation completes, the app launches automatically and prompts the customer to sign in or create an account. The Windows client requires .NET Desktop Runtime 8.0 or later, which the installer will download and install automatically if it is not already present. Corporate machines with restricted internet access may need the runtime pre-installed via their organization's software deployment tool.

3.3 Installing the macOS Application

The macOS version is distributed both through the Mac App Store and as a direct .dmg download for organizations that manage software outside the App Store. Apple Silicon (M-series) and Intel Macs are both supported through a universal binary. After downloading the .dmg, customers drag the TechMart Connect icon into the Applications folder. On first launch, macOS Gatekeeper will require the customer to right-click the app and choose 'Open' rather than double-clicking, since the app is notarized but not distributed through the App Store in the .dmg version. The app requests Local Network permission on first launch, which is required to discover Home Hub devices on the same Wi-Fi network; denying this permission will prevent device setup from completing.

3.4 Installing the Mobile Applications (iOS and Android)

The iOS app requires iOS 16 or later and is available exclusively through the Apple App Store. The Android app requires Android 10 (API level 29) or later and is available through the Google Play Store, with a sideloadable APK also offered for enterprise device management scenarios through the TechMart for Business portal. Both mobile apps request Bluetooth and Local Network permissions during first launch; Bluetooth is used only during the initial Home Hub pairing handshake and is not used for ongoing operation. Location permission is requested on Android specifically because Android requires location access to scan for nearby Bluetooth and Wi-Fi devices; TechMart does not use this permission to track customer location and this is disclosed in the in-app permission explanation screen.

3.5 TechMart Home Hub Device Setup

Step-by-step pairing process

- Plug in the Home Hub and wait for the status LED to pulse blue, indicating it is in pairing mode.
- Open the TechMart Connect app and tap 'Add Device' from the home screen.
- Select 'Home Hub' from the device category list.
- Enable Bluetooth on the phone if prompted; the app uses Bluetooth Low Energy (BLE) to transmit Wi-Fi credentials securely to the Hub.
- Choose the 2.4GHz Wi-Fi network from the list and enter the network password.
- Wait for the status LED to turn solid green, indicating the Hub has connected to Wi-Fi and registered with the TechMart cloud.
- Name the Hub (for example, 'Living Room Hub') and assign it to a room within the app.

If the status LED instead turns solid red, this indicates a failed Wi-Fi connection, most commonly due to an incorrect password, a 5GHz-only network being selected, or a router MAC address filter blocking the Hub. If the LED pulses purple, the Hub has connected to Wi-Fi but failed to reach the TechMart cloud service, which usually indicates a firewall blocking outbound traffic on port 443 or 8883 (used for the MQTT device messaging protocol).

3.6 Firmware Provisioning

New Home Hub devices ship with a base firmware image and download the latest stable firmware automatically during the first setup, provided the process is not interrupted. This initial firmware download is typically 40 to 60MB and can take two to five minutes depending on network speed. Interrupting power during a firmware update can, in rare cases, leave the device in a recovery state, indicated by a slow-blinking amber LED; recovery mode devices can be restored using the TechMart Recovery Tool available at recovery.techmart-electronics.example, which requires a direct USB-C connection to a computer.

3.7 TechMart Developer API License Activation

Developers building integrations against the TechMart Developer API must first register an application in the Developer Console at developers.techmart-electronics.example. Registration issues a client ID and client

secret used for the OAuth 2.0 client-credentials flow. Free-tier API access allows up to 1,000 requests per day across a maximum of five registered devices, suitable for testing and small personal projects. Production access requires submitting the application for review, which typically takes two to three business days, after which the daily request quota increases based on the selected plan tier (Starter, Growth, or Enterprise).

```
curl -X POST https://api.techmart-electronics.example/oauth/token \  
-d grant_type=client_credentials \  
-d client_id=YOUR_CLIENT_ID \  
-d client_secret=YOUR_CLIENT_SECRET
```

The token endpoint above returns a bearer token valid for one hour, which must be included as an Authorization header on all subsequent API requests. See Section 5 for a full reference of API error codes encountered during and after activation.

3.8 Uninstallation and Factory Reset

To fully uninstall the desktop application on Windows, use 'Add or Remove Programs' and select TechMart Connect, which also offers to remove the background sync service and local cache. On macOS, dragging the app to the Trash removes the application bundle, but customers who want to remove all local data should also delete ~/Library/Application Support/TechMart. To factory reset a Home Hub device, hold the physical reset button on the underside of the device for 10 seconds until the LED flashes white three times; this erases all local Wi-Fi credentials and unpairs the device from the customer's account, requiring the full setup process in Section 3.5 to be repeated.

3.9 Migrating from a Third-Party Smart Home Platform

Customers moving from a competing smart home platform can use the Import Wizard within the TechMart Connect mobile app, which scans the local network for compatible devices already provisioned on the same Wi-Fi network and offers to re-pair them under a TechMart account without requiring a full factory reset of the underlying hardware, provided the device manufacturer supports the open Matter smart home standard. Devices using a proprietary, non-Matter protocol must be factory reset and re-paired manually using their original manufacturer app before they can be added to TechMart Connect. The Import Wizard preserves device naming and room assignments where possible, but automation rules and schedules must be recreated manually, as these are not portable between platforms.

3.10 Offline and Low-Bandwidth Installation Considerations

In areas with limited internet bandwidth, the initial firmware download described in Section 3.6 can take considerably longer than the typical two-to-five-minute window, and customers on data-capped connections should be aware the download is approximately 40 to 60MB. For installations at sites with no reliable internet access at all, such as remote vacation properties, TechMart offers a Pre-Provisioning service through the TechMart for Business portal, in which a Hub can be fully configured and paired to a customer's account at a retail location or warehouse before shipping, requiring only a basic network connection at the final site to complete activation rather than a full first-time setup.

Section 4: Troubleshooting

Document ID: TM-TECH-TROUBLESHOOT-04 | TechMart Electronics Technical Support Knowledge Base | Category: Technical Support

This section provides general troubleshooting methodology as well as specific guidance for connectivity problems, performance issues, application crashes, and Home Hub device malfunctions.

4.1 General Troubleshooting Methodology

Support agents should follow a consistent diagnostic order for any technical issue: first confirm the scope of the problem (single device, single account, or widespread), then check the internal Status Dashboard for any known incidents, then gather basic environment information (app version, device OS version, network type), and only then move into product-specific diagnostic steps. Roughly 20% of tickets that appear to be product bugs are resolved simply by confirming the customer is running the latest application version, since older versions are sometimes affected by server-side API changes that are not backward compatible beyond a 90-day support window.

4.2 Connectivity Issues

Connectivity problems are the single largest category of Home Hub-related tickets. The most common root causes, in order of frequency, are: the Hub losing its Wi-Fi connection due to a router reboot or ISP outage, the customer's router changing Wi-Fi channels automatically (common with mesh systems performing self-optimization), the Hub being placed too far from the router causing intermittent signal drops, and band-steering features on modern routers that attempt to move the Hub to 5GHz, which it does not support for its primary connection.

Connectivity diagnostic checklist

- Check the Hub's status LED: solid green means connected, pulsing amber means reconnecting, solid red means disconnected.
- Confirm the customer's phone and Hub are on the same Wi-Fi network, not connected via cellular data and Wi-Fi respectively.
- Ask the customer to check their router admin panel for the Hub's connection status and signal strength.
- If using a mesh Wi-Fi system, recommend disabling band-steering for the Hub's specific MAC address, or creating a dedicated 2.4GHz-only network name (SSID) for IoT devices.
- Test whether other smart home devices on the same network are also experiencing drops, which points to a router-level issue rather than a Hub-specific fault.

4.3 Application Performance Issues

Customers occasionally report the TechMart Connect app becoming slow or unresponsive, particularly on households with a large number of registered devices (25 or more). This is a known scaling limitation in application versions prior to 4.8, which used a single synchronous data fetch on app launch; versions 4.8 and later implement paginated, lazy loading of device lists and have largely resolved this complaint. Agents

should recommend updating to the latest version before pursuing deeper diagnostics. On mobile devices, performance issues are also frequently linked to low available storage, since the app caches device thumbnails and event history locally; clearing the app cache from Settings > Storage > TechMart Connect > Clear Cache on Android, or offloading and reinstalling the app on iOS, typically resolves this.

4.4 Application Crashes and Freezes

When an app crashes repeatedly, agents should request a diagnostic log bundle, which customers can generate from Settings > Help > Send Diagnostic Report within the app. This bundle includes recent crash logs, device information, and a redacted copy of the local event cache, and is automatically attached to the support ticket for engineering review. Common crash triggers identified in past incident reviews include corrupted local cache files after an interrupted app update, conflicts with certain third-party VPN applications that intercept local network traffic, and, on older Android devices with limited RAM, out-of-memory crashes when more than 40 devices are registered to a single account.

Steps before escalating a crash ticket

- Confirm the app is fully updated to the latest available version for the platform.
- Ask the customer to restart their device, not just the app, to rule out a temporary OS-level memory issue.
- Request the diagnostic report bundle described above.
- Check whether the crash is reproducible with a specific action (e.g., always crashes when opening the Energy Usage tab) versus random, as this significantly narrows the root cause.

4.5 Home Hub Device Malfunctions

Physical device issues are diagnosed primarily through the Hub's LED status patterns, summarized in the table below. Agents should always ask the customer to describe the exact LED color and pattern (solid, slow pulse, fast pulse, or blink count) rather than assuming based on a general description of 'it's not working.'

LED Pattern	Meaning	Recommended Action
Solid green	Connected and operating normally	No action needed
Pulsing blue	In pairing mode	Proceed with setup in Section 3.5
Solid red	Wi-Fi connection failed	Check network credentials and 2.4GHz availability
Pulsing purple	Connected to Wi-Fi, cannot reach cloud	Check firewall rules on ports 443 and 8883
Slow amber blink	Firmware recovery mode	Use the TechMart Recovery Tool
Fast white blink	Overheating protection engaged	Move device to a well-ventilated location
No light at all	Power failure or hardware fault	Test a different outlet and USB-C cable before RMA

4.6 Audio and Voice Assistant Integration Issues

Customers who connect TechMart Home Hub to third-party voice assistants occasionally report that voice commands work intermittently. Because voice assistant integrations rely on TechMart's cloud-to-cloud

skill/action API rather than direct local communication, latency or brief outages on either the voice platform side or the TechMart side can cause commands to silently fail. Agents should confirm whether the issue occurs with the manufacturer's own app (indicating a TechMart-side issue) or only through the voice assistant (indicating a skill/action linking issue), and can recommend the customer unlink and relink the TechMart skill from within the voice assistant's companion app as a first step.

4.7 Data Sync and Multi-User Household Issues

In households where multiple family members share access to the same Home Hub devices through the Family Sharing feature, delays in state synchronization (for example, one family member turning off a light but another family member's app still showing it as on) are almost always caused by that second device being in a poor network state rather than a bug in the sharing feature itself, since state updates are pushed via a persistent MQTT connection that requires a stable network connection to receive in real time. Asking the affected family member to pull-to-refresh the device list, or to fully close and reopen the app, forces a fresh state fetch and typically resolves the discrepancy immediately.

4.8 Battery-Powered Sensor Troubleshooting

TechMart battery-powered accessories, such as door and window sensors that pair with the Home Hub, communicate using a low-power variant of the local mesh protocol rather than Wi-Fi directly, and report a low-battery warning in the app once voltage drops below the manufacturer-defined threshold, typically corresponding to two to three weeks of remaining life under normal use. Customers reporting a sensor that has 'stopped working' entirely, rather than showing a low-battery warning, most commonly have a battery that has already fully drained past the point where the sensor could transmit a final low-battery notification; replacing the battery with a fresh one of the same type (typically CR2032) resolves the vast majority of these reports. Sensors that remain unresponsive after a battery replacement should be re-paired using the 'Replace Device' option in the app rather than 'Add New Device,' which preserves the sensor's existing automation rules.

4.9 Interpreting the Internal Status Dashboard

The internal Status Dashboard, updated automatically from infrastructure health checks, displays real-time status for each regional cluster, the MQTT messaging backbone, the Developer API, and push notification delivery. A yellow 'Degraded Performance' status for a component indicates elevated error rates or latency that may explain a spike in similar tickets without representing a full outage, while a red 'Service Disruption' status confirms a known, actively-being-fixed issue and should be referenced directly in customer communications rather than customers being asked to continue troubleshooting on their end. Agents should check this dashboard at the start of any shift and before spending significant time on an issue that appears to affect multiple, unrelated customers simultaneously.

Section 5: API Errors

Document ID: TM-TECH-APIERR-05 | TechMart Electronics Technical Support Knowledge Base | Category: Technical Support

This section documents the TechMart Developer API error taxonomy for third-party developers and internal support agents assisting Developer Console customers. All API errors are returned as JSON objects with an 'error_code', 'message', and optional 'details' field, alongside the corresponding HTTP status code.

```
{
  "error_code": "AUTH_TOKEN_EXPIRED",
  "message": "The provided access token has expired.",
  "details": { "expired_at": "2026-06-30T14:02:11Z" }
}
```

5.1 Authentication and Authorization Errors (4xx)

HTTP	error_code	Meaning
401	AUTH_TOKEN_MISSING	No Authorization header was provided on the request
401	AUTH_TOKEN_EXPIRED	Access token has passed its one-hour expiry
401	AUTH_TOKEN_INVALID	Token signature could not be verified
403	AUTH_SCOPE_INSUFFICIENT	Token lacks the required scope for this endpoint
403	AUTH_APP_SUSPENDED	Developer application has been suspended for policy violations
429	RATE_LIMIT_EXCEEDED	Daily or per-minute request quota has been exceeded

AUTH_TOKEN_EXPIRED is by far the most common integration error reported by developers new to the platform, typically caused by caching a token for longer than its one-hour lifetime instead of requesting a fresh token or implementing the refresh flow. Developers should implement automatic token renewal at roughly the 50-minute mark rather than waiting for a 401 response, to avoid dropped requests during renewal.

5.2 Rate Limiting

The API enforces both a per-minute burst limit and a daily quota. The per-minute limit is 60 requests per client for the free tier and 300 requests per client for paid tiers. When a request is rate-limited, the response includes a Retry-After header indicating the number of seconds to wait before retrying. Developers are strongly encouraged to implement exponential backoff rather than retrying immediately, since immediate retries during a rate-limit window will simply extend the penalty period under the platform's abuse-prevention logic. Sustained abuse of the rate limit, defined as exceeding the limit more than 50 times in a 24-hour period, can result in a temporary application suspension (AUTH_APP_SUSPENDED) requiring a manual review request through the Developer Console.

5.3 Device Command Errors (5xx and 4xx)

HTTP	error_code	Meaning
------	------------	---------

404	DEVICE_NOT_FOUND	Device ID does not exist or is not linked to this application
409	DEVICE_OFFLINE	Command could not be delivered because the device is not connected
422	COMMAND_UNSUPPORTED	The requested command is not supported by this device model
500	COMMAND_TIMEOUT	Device did not acknowledge the command within 10 seconds
503	REGION_UNAVAILABLE	The device's regional cluster is undergoing maintenance

DEVICE_OFFLINE is a normal and expected response rather than a bug; developers should design their integrations to queue commands and retry, or clearly surface an offline state to the end user, rather than treating it as an unrecoverable error. COMMAND_TIMEOUT is more concerning if it occurs repeatedly for the same device, and often correlates with the connectivity issues described in Section 4.2.

5.4 Webhook Delivery Errors

Developers can subscribe to real-time device events via webhooks configured in the Developer Console. If a webhook endpoint returns a non-2xx response, TechMart retries delivery up to five times with exponential backoff over a 15-minute window, after which the event is marked as failed and logged in the Webhook Delivery Log. If a webhook endpoint fails more than 500 consecutive deliveries, the subscription is automatically disabled (error code WEBHOOK_AUTO_DISABLED) to prevent unnecessary load on both systems, and must be manually re-enabled from the Developer Console after the underlying issue is fixed. All webhook payloads are signed using HMAC-SHA256 with the application's webhook secret; a WEBHOOK_SIGNATURE_INVALID error in the delivery log indicates the receiving endpoint is either using the wrong secret or has a bug in its signature verification logic, not a problem on TechMart's side.

5.5 SDK-Specific Errors

TechMart provides official SDKs for Python, Node.js, and Java. A common source of confusion is that the Python SDK versions prior to 2.0 raised a generic TechMartAPIError for all failure types, while version 2.0 and later raise typed exceptions (TechMartAuthError, TechMartRateLimitError, TechMartDeviceError) that map directly to the error categories above. Developers reporting unexpected exception types should be asked which SDK version they are using and directed to upgrade, since the untyped errors in 1.x made granular error handling difficult and are no longer actively maintained.

```
try:
    client.devices.send_command(device_id, "turn_on")
except TechMartDeviceError as e:
    if e.error_code == "DEVICE_OFFLINE":
        queue_for_retry(device_id)
    else:
        raise
```

5.6 Sandbox vs. Production Environment Errors

Developers frequently mix up sandbox and production API base URLs, resulting in confusing errors such as DEVICE_NOT_FOUND for a device that clearly exists. The sandbox environment (sandbox-api.techmart-electronics.example) is a fully isolated environment with simulated virtual devices and does not share any data with the production environment (api.techmart-electronics.example). Support

agents should always confirm the base URL in use when a developer reports a data-related error that does not match what they see in their own account.

5.7 Validation and Input Errors

HTTP	error_code	Meaning
400	VALIDATION_MISSING_FIELD	A required field was omitted from the request body
400	VALIDATION_INVALID_TYPE	A field was provided with an unexpected data type
400	VALIDATION_OUT_OF_RANGE	A numeric value fell outside the accepted range for that field
413	PAYLOAD_TOO_LARGE	Request body exceeded the 1MB maximum size limit

Validation errors include a 'details' object listing every offending field in a single response rather than only the first error encountered, which allows developers to fix all issues in one iteration instead of a slow trial-and-error cycle. Developers integrating against the API for the first time are encouraged to validate request payloads against the published OpenAPI schema in the Developer Console before sending live requests, which catches the majority of VALIDATION_ errors during local development.

5.8 Deprecation and Versioning Errors

The API uses a date-based versioning scheme specified via an X-API-Version header; requests that omit this header default to the oldest still-supported version rather than the latest, which can cause confusing behavior for new integrations that expect current default behavior. When an API version is formally deprecated, requests continue to function for a minimum 12-month sunset window but return a Deprecation-Warning response header pointing to migration documentation. Requests made after the sunset date return error code API_VERSION_RETIRED with a 410 Gone status, and developers must update their integration to a supported version, as retired versions cannot be re-enabled even temporarily.

Section 6: System Requirements

Document ID: TM-TECH-SYSREQ-06 | TechMart Electronics Technical Support Knowledge Base | Category: Technical Support

This section lists the minimum and recommended system requirements for all TechMart Connect client applications, the Home Hub device family, and network prerequisites required for reliable operation.

6.1 Windows Desktop Application

Component	Minimum	Recommended
Operating System	Windows 10, version 21H2 or later	Windows 11, latest update
Processor	Dual-core 1.6GHz	Quad-core 2.0GHz or better
Memory (RAM)	4GB	8GB or more
Storage	500MB free disk space	1GB free disk space
Network	Broadband internet connection	Broadband, low-latency, Wi-Fi 5 or better router
Runtime	.NET Desktop Runtime 8.0	.NET Desktop Runtime 8.0 (auto-updated)

6.2 macOS Application

Component	Minimum	Recommended
Operating System	macOS 13 Ventura	macOS 15 Sequoia or later
Processor	Apple M1 or Intel Core i5 (8th gen)	Apple M2 or later
Memory (RAM)	8GB	16GB
Storage	500MB free disk space	1GB free disk space

6.3 Mobile Applications

Platform	Minimum OS	Recommended	Storage
iOS	iOS 16.0	iOS 18 or later	250MB free
Android	Android 10 (API 29)	Android 13 or later	250MB free

Older devices that still meet the minimum OS version but have less than 2GB of total RAM may experience the performance issues described in Section 4.3. iPad devices are supported using the iPhone app in compatibility mode; a dedicated iPad layout is planned but not yet available as of this document's version.

6.4 TechMart Home Hub Hardware Specifications

Specification	Home Hub (Gen 3)	Home Hub Mini
Wi-Fi	802.11 b/g/n, 2.4GHz only	802.11 b/g/n, 2.4GHz only

Bluetooth	Bluetooth 5.2 LE (setup only)	Bluetooth 5.0 LE (setup only)
Power	USB-C, 9W adapter included	USB-C, 5W adapter included
Max connected devices	150 per Hub	50 per Hub
Operating temperature	0 to 40 degrees C	0 to 40 degrees C
Local storage	8GB (event history buffer)	2GB (event history buffer)

It is important to note that both Home Hub models operate exclusively on the 2.4GHz Wi-Fi band and do not support 5GHz or 6GHz networks, even though many customer routers broadcast a single combined SSID across all bands. Customers with combined-band routers experiencing setup failures should be guided to temporarily create a separate 2.4GHz-only network name for the pairing process, as described in Section 3.5.

6.5 Network and Firewall Requirements

For reliable Home Hub operation, the following outbound connections must be permitted on the customer's network. Agents supporting business or advanced home network customers with strict firewalls should share this list directly.

Purpose	Host	Port	Protocol
Cloud API and app sync	api.techmart-electronics.example	443	HTTPS
Device messaging (MQTT)	mqtt.techmart-electronics.example	8883	MQTTS
Firmware updates	fw.techmart-electronics.example	443	HTTPS
Time synchronization	ntp.techmart-electronics.example	123	NTP/UDP
Diagnostics upload	diag.techmart-electronics.example	443	HTTPS

Networks that use deep packet inspection (DPI) or SSL-inspecting proxies may interfere with certificate pinning used on the MQTT connection, resulting in the pulsing purple LED state described in Section 4.5. Corporate and school networks are the most common source of these issues; agents should ask whether the affected device is on a managed network before spending significant time on consumer router troubleshooting.

6.6 TechMart Developer API Requirements

The Developer API is a standard REST/JSON API reachable over HTTPS (TLS 1.2 or later) with no special client requirements beyond the ability to make outbound HTTPS requests and handle JSON. Official SDKs require Python 3.9 or later, Node.js 18 or later, or Java 11 or later. Webhook receiving endpoints must respond within 10 seconds and must be reachable over HTTPS with a valid, non-self-signed TLS certificate; self-signed certificates will cause all webhook deliveries to fail silently from the developer's perspective, appearing in the Webhook Delivery Log as WEBHOOK_SIGNATURE_INVALID or a generic connection failure.

6.7 Browser Requirements for the Web Portal

Browser	Minimum Version	Notes
Google Chrome	Last 2 major versions	Fully supported, recommended
Mozilla Firefox	Last 2 major versions	Fully supported
Microsoft Edge	Last 2 major versions	Fully supported (Chromium-based only)
Safari	Version 16 or later	See Section 1.7 for ITP-related login notes
Internet Explorer	Not supported	Web portal will display an unsupported-browser notice

6.8 End-of-Support Policy

TechMart maintains active support for the two most recent major versions of each client application. Versions older than 90 days past a major release are considered end-of-life and may not be compatible with current API responses, particularly as new device types and features are introduced. Support agents should always confirm the application version at the start of any technical ticket, since a significant share of reported bugs are already resolved in the current release and simply require the customer to update.

6.9 Accessibility Requirements and Support

All TechMart Connect client applications are developed to conform to WCAG 2.1 Level AA guidelines, including full screen-reader support on iOS (VoiceOver), Android (TalkBack), and the web portal, as well as complete keyboard navigation on desktop and web without requiring a mouse. High-contrast mode and adjustable text sizing follow the operating system's own accessibility settings automatically rather than requiring a separate in-app configuration. Customers reporting accessibility issues, such as an unlabeled button or a color-contrast concern, should have their ticket tagged Accessibility and routed to the Product Accessibility team, which maintains a separate, expedited review process distinct from standard feature-request tickets.

6.10 Storage and Bandwidth Considerations for Video Accessories

TechMart video-capable accessories, such as the optional Home Hub Doorbell Camera, require additional bandwidth beyond the baseline Hub requirements: a sustained upload speed of at least 2 Mbps per active camera is recommended for smooth live-view streaming, and considerably more if multiple cameras are viewed simultaneously by different family members. Recorded event clips are stored in the cloud for a rolling 14-day period on the free tier or 30 days on a paid Video History subscription, and local storage is not supported on any current TechMart camera model, meaning a camera that loses internet connectivity will not record events locally during the outage, only resuming cloud recording once connectivity is restored.

Appendix

Extended Frequently Asked Questions by Category

The following appendix expands on Sections 1 through 6 with additional real-world scenarios encountered by the TechMart support team. Entries are organized by category and are intended to supplement, not replace, the primary diagnostic steps described earlier in this document.

Appendix A: Login Issues — Extended FAQ

Document ID: TM-TECH-LOGIN-FAQ-A1 | TechMart Electronics Technical Support Knowledge Base | Category: Technical Support

This appendix addresses less common but recurring login scenarios reported through support tickets, chat transcripts, and community forum threads over the past year.

Why does the app say my account exists but my password never works, even right after a reset?

This typically happens when the customer has two TechMart accounts under similar email addresses, such as a personal Gmail address and a Google Workspace address that auto-fills. Ask the customer to confirm the exact email shown on their most recent order confirmation email before resetting again.

A customer says they can log into the mobile app but not the web portal with the same credentials. What's happening?

The mobile app and web portal share the same authentication backend, so this usually indicates a browser-level issue rather than an account issue. Have the customer clear cookies for techmart-electronics.example or try a different browser before escalating.

Can a customer be logged into more than five devices at once?

Yes. TechMart Connect allows unlimited concurrent sessions on personal accounts, though TechMart for Business accounts can be configured by an administrator to cap concurrent sessions per user for security compliance purposes.

What happens to active sessions when a customer changes their email address?

Changing the primary email address does not automatically revoke existing sessions, but it does require the customer to verify the new address via a confirmation link within 24 hours, after which the old address can no longer be used to initiate password resets.

A customer's login works fine on Wi-Fi but fails on cellular data. Why?

This is almost always caused by a carrier-level DNS filter or a corporate mobile device management (MDM) profile restricting access to non-approved domains. Ask the customer to try a different DNS provider or temporarily disable any MDM profile to confirm.

Is there a way to see a customer's login history without giving them account access?

Yes, the Security Timeline in the Account Administration Console shows the last 90 days of login attempts, including IP address, approximate location, device type, and success or failure status, and is visible to agents after identity verification.

Why do some customers see a 'this browser is not supported' message even on Chrome?

This message appears when the browser's user-agent string has been modified by an extension or when the customer is running Chrome in a restricted enterprise mode that disables modern JavaScript features

required by the login form.

Do login sessions expire faster on shared or public computers?

Customers can choose 'This is a public or shared computer' at login, which shortens the session to 15 minutes of inactivity instead of the standard 30-day refresh token window, and disables the 'remember this device' option for MFA.

What should an agent do if a customer reports their account was accessed from a country they've never visited?

Immediately force a password reset, revoke all active sessions from the Security Timeline, and recommend enabling MFA if it is not already active. This should be treated as a possible compromise regardless of whether the customer noticed any changes to their devices.

Why does SSO login sometimes create a duplicate account instead of linking to an existing one?

This occurs when the email address returned by the identity provider does not exactly match the case or formatting of the email on the existing account, for example due to plus-addressing (name+tag@example.com). The Account Merge tool in the Administration Console can combine the two accounts after identity verification.

A customer reports the login page itself never finishes loading, showing only a blank white screen. What should be checked?

A persistently blank login screen is most often caused by a corporate content filter blocking one of the static asset domains used by the web portal rather than the primary domain itself. Ask the customer to open their browser's developer console and check for network errors referencing cdn.techmart-electronics.example, which is separate from the primary auth domain and sometimes missed by manually configured allowlists.

Can a customer disable SSO after initially setting up their account with Google or Apple sign-in?

Yes, customers can add a traditional password to an SSO-based account from Account Settings > Security > Set Password, after which both the original SSO method and the new password can be used interchangeably to sign in, rather than one replacing the other.

Why would a customer be prompted for MFA on a device they use every day?

Devices are only remembered for MFA purposes for 30 days from the last successful MFA challenge; a customer who has not opened the app in over a month, or who cleared their app data, will be prompted again even though the device itself is otherwise recognized.

Appendix B: Password Reset — Extended FAQ

Document ID: TM-TECH-PWRESET-FAQ-B1 | TechMart Electronics Technical Support Knowledge Base | Category: Technical Support

This appendix covers edge cases in the password reset and account recovery flow that fall outside the standard self-service process.

A customer's reset link says it's expired even though they clicked it within a minute of receiving it. Why?

Some corporate email security gateways 'pre-click' links in incoming emails to scan them for phishing before delivering the message to the inbox, which consumes the single-use reset link before the customer ever sees it. Recommend the customer's IT department exclude TechMart's sending domain from link-scanning, or use the 'resend as plain text' option available on the reset page.

Can a customer reset their password from an old, deauthorized device?

Yes, the password reset flow only requires access to the registered email address and does not require an authenticated device, since it is specifically designed for scenarios where the customer no longer trusts or has access to any signed-in device.

What is the difference between 'Sign out of all devices' and a password reset?

'Sign out of all devices' revokes active session tokens immediately but keeps the existing password unchanged, while a password reset both changes the password and, as a side effect, also revokes all active sessions.

Why does the password policy reject a password that looks strong, like 'Sunshine2026!'?

This password would be rejected by the breach-database check (PWD-010) because it follows an extremely common pattern (capitalized word, year, symbol) that appears frequently in leaked password lists, even though it satisfies the basic complexity rules.

Is there a way to reset a password without triggering an email at all, for testing purposes?

No. All password resets, including those initiated from the internal Account Administration Console, send a notification email to the customer for security and audit purposes; this cannot be suppressed, even in test or sandbox environments.

A business customer wants to require MFA for all employees. Where is that configured?

Organization administrators can enforce mandatory MFA from the Admin Console under Organization > Security > Require Multi-Factor Authentication, which prevents any user in the organization from disabling MFA on their individual account.

How long does a customer have before they must set a new password after an agent-assisted reset is initiated?

The reset link sent as part of an agent-assisted reset uses the same 20-minute expiry as the standard self-service flow; there is no extended window for agent-initiated resets.

What should an agent do if a customer insists they never requested a password reset email they received?

This can be a legitimate customer testing whether an account exists, or in rare cases an early sign of an account-enumeration attempt. No action is needed if the customer did not click the link, but agents should note the ticket and monitor the Security Timeline for further suspicious activity.

Does resetting a password affect a customer's saved payment methods or subscription?

No, a password reset only affects the credential used to sign in and has no impact on billing information, active subscriptions, or connected devices, all of which remain unchanged and fully intact.

Can a customer set the same password back to what it was before, immediately after a forced reset?

No, the last-five-passwords check described in Section 2.2 (PWD-011) prevents reusing the immediately preceding password, which is intentional so that a forced reset following a suspected compromise cannot simply be undone by the customer setting the same, potentially already-known password again.

What should an agent tell a customer who wants to know why the password reset link can only be used once?

The single-use design prevents the link from being replayed if it is later found in a shared inbox, browser history, or forwarded email; explaining this rationale briefly tends to reduce follow-up frustration when a customer accidentally clicks an already-used link a second time.

Appendix C: Installation — Extended FAQ

Document ID: TM-TECH-INSTALL-FAQ-C1 | TechMart Electronics Technical Support Knowledge Base | Category: Technical Support

This appendix answers additional installation questions related to enterprise deployment, offline environments, and less common hardware configurations.

Can the Windows app be deployed silently across an organization?

Yes, the installer supports a silent flag: TechMartConnectSetup.exe /S /norestart, which is commonly deployed through Group Policy or an RMM tool. A separate MSI package is also available on request from the TechMart for Business portal for organizations that require MSI-based deployment.

Does the Home Hub require an active internet connection to function at all, or does it work locally?

Basic on/off and scheduling commands issued from a device on the same local network continue to function during an internet outage using local MQTT communication, but remote access, voice assistant integration, and automation rules that depend on cloud-based triggers such as weather or sunset time will not function until connectivity is restored.

Can two Home Hubs be set up in the same household?

Yes, TechMart supports multiple Hubs per household, which is recommended for larger homes to improve Bluetooth pairing range and reduce the number of devices connected to a single Hub. Each Hub is assigned to a room within the app but all Hubs share the same household account.

Is a factory reset required to transfer a Home Hub to a new owner?

Yes, a full factory reset as described in Section 3.8 is required, after which the new owner can add the device to their own account. TechMart does not support direct account-to-account device transfer to prevent accidental loss of device history.

What happens if the Wi-Fi password is changed after the Hub is already set up?

The Hub will disconnect and its LED will turn solid red. Customers do not need to fully reset the device; instead, they can use 'Update Wi-Fi Network' from the device settings menu within the app, which re-triggers the Bluetooth pairing handshake without erasing other device configurations.

Can the Home Hub be set up without a smartphone, using only the web portal?

No, initial pairing requires the Bluetooth Low Energy handshake performed by the mobile app; the web portal can manage devices after setup is complete but cannot perform the initial pairing step.

Is the Developer API sandbox environment reset periodically?

Yes, sandbox environment data, including virtual devices and webhook logs, is automatically purged every 30 days to keep the test environment manageable; developers relying on long-lived sandbox test data should account for this in their testing workflows.

What is the recommended distance between the Home Hub and the router?

TechMart recommends no more than 30 feet (approximately 9 meters) with no more than two interior walls between the Hub and the router for reliable 2.4GHz signal strength; homes with thick walls, metal framing, or multiple floors may require a Wi-Fi extender or mesh node near the Hub.

Is a wired Ethernet connection an option for the Home Hub instead of Wi-Fi?

The Home Hub Gen 3 includes an Ethernet port that, when connected, automatically takes priority over Wi-Fi and generally provides more stable performance for large households; the Home Hub Mini does not include an Ethernet port and is Wi-Fi only.

Can the mobile app be used to set up a Hub while connected to a guest Wi-Fi network?

No, guest networks typically isolate connected clients from each other for security reasons, which prevents the phone from discovering the Hub over the local network during the Bluetooth-to-Wi-Fi handoff; the phone and Hub must both be set up using the primary household network, at least temporarily.

What should be done if the setup code on the Home Hub's label is smudged or unreadable?

The setup code can also be found printed inside the original retail packaging and is duplicated in the Account Administration Console once the device's serial number is looked up, provided the customer purchased directly from TechMart or a registered retailer with a valid order number.

Appendix D: Troubleshooting — Extended FAQ

Document ID: TM-TECH-TROUBLESHOOT-FAQ-D1 | TechMart Electronics Technical Support Knowledge Base | Category: Technical Support

This appendix captures troubleshooting scenarios that fall outside the standard diagnostic flows but appear repeatedly in support ticket analysis.

A customer reports their automation schedules run a few minutes late every day. Why?

Scheduled automations are evaluated by the cloud scheduler in one-minute intervals and can experience up to a two-minute delay during periods of high regional load; this is expected behavior and not a bug, though delays consistently longer than five minutes should be escalated with the affected region and timestamp.

Why does energy usage data sometimes show a gap of a few hours in the app's history graph?

Energy usage data is buffered locally on the Hub and uploaded in batches every 15 minutes; if the Hub loses internet connectivity during that window, the buffered data for that period is discarded once the local 8GB buffer rotates, rather than being resent, which can create small gaps during extended outages.

A customer's device shows as 'online' in the app but does not respond to commands. What should be checked first?

This 'ghost online' state usually means the Hub has an active connection to the cloud's presence service but its MQTT command channel has silently dropped; asking the customer to toggle Wi-Fi off and on on their router, or power-cycling the Hub, typically restores the command channel faster than waiting for the automatic 10-minute reconnect check.

Why do notifications sometimes arrive late on iOS but not Android?

iOS more aggressively suspends background app refresh to conserve battery, which can delay push notification delivery by several minutes under Apple's silent push throttling; advising customers to disable Low Power Mode and enable Background App Refresh for TechMart Connect in iOS Settings typically resolves this.

A customer says the app shows a device in the wrong room after a recent update. Is this a bug?

This has been a known display-only issue in versions 4.6 and 4.7 where room assignment order could shuffle after a sync, without actually changing the underlying room assignment in the database; updating to 4.8 or later resolves the display issue and no data is lost.

What is the correct process for a customer who suspects their Hub was damaged by a power surge?

Confirm the LED shows no light at all even when plugged into a known-working outlet with the original cable, then check the Warranty status in the Account Administration Console; power-surge damage within the standard one-year warranty period is covered and should be processed as a hardware RMA rather than continued software troubleshooting.

Why would a customer see duplicate event history entries for the same action?

Duplicate entries can occur when a command is issued from two different clients nearly simultaneously, such as a physical wall switch and the app, both of which independently log the resulting state change; this is expected logging behavior rather than a sync bug.

A customer's app shows a device as 'updating' indefinitely. When should this be escalated?

Firmware updates for connected accessories typically complete within five minutes; if the 'updating' state persists beyond 20 minutes, the update process has likely stalled and the affected accessory should be power-cycled, which safely resumes the update from where it left off in the vast majority of TechMart accessory models.

Why do some customers report the app's temperature readings differ slightly from a separate thermometer?

TechMart climate sensors are factory-calibrated to within plus or minus 1 degree Fahrenheit, and small discrepancies against an uncalibrated household thermometer are expected; readings differing by more than 3 degrees consistently should be reported as a possible sensor calibration issue eligible for warranty replacement.

What is the first thing to check when a customer reports that the entire app appears to be in the wrong language?

The app language follows the device's system-level language setting by default rather than a separate in-app setting; confirming and, if needed, correcting the phone or computer's system language resolves this in almost all cases, since TechMart does not currently offer an independent in-app language override.

Appendix E: API Errors — Extended FAQ

Document ID: TM-TECH-APIERR-FAQ-E1 | TechMart Electronics Technical Support Knowledge Base | Category: Technical Support

This appendix addresses developer questions frequently raised in the Developer Console support queue that go beyond the core error code reference in Section 5.

Why do I get AUTH_TOKEN_INVALID immediately after successfully requesting a token?

This is almost always caused by copying the token with extra whitespace or a truncated string due to a logging or console line-wrap issue; verify the Authorization header is exactly 'Bearer ' followed by the unmodified token with no line breaks.

Does the API support pagination for accounts with many devices?

Yes, the /v1/devices endpoint supports cursor-based pagination using a 'next_cursor' field in the response; developers who fetch all devices in a single unpaginated call on large accounts will encounter unexpectedly truncated results rather than an explicit error, which is a common source of confusion.

Why does a webhook that works fine in Postman fail when triggered by a real event?

This is typically caused by testing the endpoint manually with a valid signature header copied from a previous request; real webhook deliveries generate a new HMAC signature for every payload, so any endpoint that hardcodes or caches a signature value for validation will always reject real events.

Is there a sandbox equivalent for testing rate-limit handling?

Yes, the sandbox environment enforces the same per-minute and daily limits as production, so developers can safely test their backoff and retry logic against sandbox-api.techmart-electronics.example without risking their production application's standing.

What is the correct way to handle a COMMAND_TIMEOUT versus a DEVICE_OFFLINE error?

DEVICE_OFFLINE means the platform already knows the device has no active connection and the command was never attempted; COMMAND_TIMEOUT means the device appeared online but did not acknowledge the command within 10 seconds, which more often indicates a local network issue on the customer's end and may resolve on a single retry.

Can a developer request a higher per-minute rate limit without upgrading their plan tier?

Temporary rate-limit increases can be requested through the Developer Console for time-limited events such as product launches, but permanent increases beyond the plan tier's stated limit require upgrading to the next tier.

Why does the API return slightly different field names in sandbox versus production for the same endpoint?

This should never happen and indicates the developer is using an outdated sandbox SDK client pinned to a previous API version; sandbox and production run the same API version at all times, and any discrepancy should be reported to Platform Engineering rather than worked around.

Is there a way to test webhook delivery without a public HTTPS endpoint during local development?

Yes, the Developer Console includes a built-in Webhook Inspector that generates a temporary, publicly reachable forwarding URL and displays incoming payloads in real time, intended specifically for local development before a permanent production endpoint is available.

Why do bulk device command requests sometimes partially succeed?

The batch command endpoint processes each device independently and returns a per-device status array rather than a single pass/fail result for the entire batch; developers must check each entry in the response array individually rather than assuming a 200 status code means every command succeeded.

What is the recommended way to detect that a developer's own token has been revoked due to a customer disconnecting the integration?

Rather than only waiting for an AUTH_TOKEN_INVALID error on the next request, developers should subscribe to the account.disconnected webhook event, which fires immediately when a customer revokes third-party access from their TechMart Connect account settings, allowing the integration to clean up gracefully instead of failing silently on the next scheduled call.

Appendix F: System Requirements — Extended FAQ

Document ID: TM-TECH-SYSREQ-FAQ-F1 | TechMart Electronics Technical Support Knowledge Base | Category: Technical Support

This appendix addresses compatibility questions that fall outside the standard requirements tables in Section 6.

Will the Home Hub work with a Wi-Fi 6E or Wi-Fi 7 router?

Yes, the Home Hub connects using standard 802.11 b/g/n on the 2.4GHz band, which remains backward compatible on all newer router standards; customers only need to ensure the router broadcasts a discoverable 2.4GHz network, since some Wi-Fi 6E routers disable 2.4GHz by default in favor of the newer bands.

Can the desktop app run inside a virtual machine?

The Windows and macOS apps are not officially tested or supported inside virtual machines, and customers running the app in a VM may experience the clock-drift session issues described in Section 1.4 more frequently due to VM time synchronization quirks.

Is there a Linux version of the desktop application?

There is no native Linux desktop client; Linux users are directed to the web portal, which supports all core account and device management features aside from local network device discovery during initial setup, which still requires the mobile app.

What happens when a customer's device falls below the minimum supported OS version after an update cycle?

The app will continue to function but will display a persistent in-app banner recommending an OS update, and new features released after that point may not be available; critical security patches to the app itself are still delivered as long as the OS can run the minimum supported app version.

Does the API enforce a minimum TLS version?

Yes, TLS 1.2 is the minimum supported version and TLS 1.0 and 1.1 connections are rejected outright at the load balancer level with a connection reset rather than an application-level error, which can be confusing for developers using very old HTTP client libraries.

Are there different Home Hub hardware requirements for the Family Sharing feature described in Section 4.7?

No, Family Sharing is a software and account-permissions feature with no additional hardware requirements beyond a standard Home Hub or Home Hub Mini already connected to the household's account.

What internet speed is recommended for a household running multiple Home Hubs and cameras together?

TechMart recommends a minimum of 25 Mbps download and 10 Mbps upload for a typical household running one Hub and up to two cameras, scaling upward for additional camera accessories as described in Section 6.10, though the Hub's own control-plane traffic is lightweight and rarely exceeds 1 Mbps on its own.

Does the Home Hub support IPv6-only networks?

The Home Hub requires IPv4 connectivity for its cloud communication and does not currently support IPv6-only networks; customers on an IPv6-only ISP configuration must ensure their router provides IPv4 connectivity via NAT64 or a dual-stack configuration.

Is a static IP address required for the Home Hub to function reliably?

No, the Hub is designed to work with standard DHCP-assigned addresses and will automatically reconnect using its existing credentials if its IP address changes; a static or reserved DHCP lease is only recommended for advanced users who also want predictable local network access to the Hub for troubleshooting purposes.

Appendix G: Glossary of Terms

Document ID: TM-TECH-GLOSSARY-G1 | TechMart Electronics Technical Support Knowledge Base | Category: Technical Support

This glossary defines terminology used throughout the TechMart Electronics Technical Support Knowledge Base. Agents should familiarize themselves with these terms, as customers and internal documentation frequently use them without further explanation.

Access Token

A short-lived credential, valid for 60 minutes, issued after successful login and used to authorize requests to the TechMart API and mobile/desktop applications.

Account Administration Console

The internal tool used by TechMart support agents to view and manage customer accounts, sessions, devices, and billing after identity verification.

Band-Steering

A router feature that automatically moves connected devices between 2.4GHz and 5GHz bands; can interfere with Home Hub connectivity since the Hub only supports 2.4GHz.

Bluetooth Low Energy (BLE)

A low-power Bluetooth protocol used exclusively during the initial Home Hub pairing process to transmit Wi-Fi credentials from the mobile app to the device.

Client Credentials Flow

An OAuth 2.0 authentication flow used by the TechMart Developer API in which an application authenticates using its client ID and client secret rather than a user's personal login.

Cluster (Regional)

One of three geographically separated data-storage and processing regions (Americas, Europe, Asia-Pacific) that a TechMart Connect account is permanently assigned to at signup.

Data Residency Migration

The internal, engineering-owned process for moving a customer account between regional clusters; not available to support agents through self-service tools.

Developer Console

The web-based portal at developers.techmart-electronics.example where third-party developers register applications, manage API credentials, and monitor usage and webhook delivery.

Diagnostic Report Bundle

A package of crash logs, device information, and redacted event history generated from within the app and automatically attached to a support ticket.

Entra ID

Microsoft's cloud identity platform (formerly Azure Active Directory), relevant to SSO login failures for customers using a Microsoft work or school account.

Family Sharing

A TechMart Connect feature allowing multiple household members to access and control the same set of registered Home Hub devices under shared or individual permission levels.

Firmware Recovery Mode

A low-level state entered by a Home Hub device when a firmware update is interrupted, indicated by a slow amber LED blink, recoverable only via a direct USB-C connection and the TechMart Recovery Tool.

Home Hub

TechMart's flagship smart home controller device, available in standard (Gen 3) and Mini form factors, responsible for local device communication and cloud connectivity.

HMAC Signature

A cryptographic signature attached to every webhook payload, generated using the developer application's webhook secret, used by the receiving endpoint to verify the payload genuinely originated from TechMart.

Identity Recovery

The manual, Trust and Safety-reviewed process for restoring account access to customers who have lost both their registered email and their MFA device.

MQTT

A lightweight publish-subscribe messaging protocol used for real-time communication between Home Hub devices and the TechMart cloud, operating over port 8883 with TLS encryption (MQTTTS).

Multi-Factor Authentication (MFA)

An additional login verification step beyond a password, supported via time-based one-time passwords (TOTP), SMS codes, or single-use backup codes.

Refresh Token

A long-lived credential, valid for 30 days, stored on the client device and used to silently obtain new access tokens without requiring the customer to log in again.

Security Timeline

A log within the Account Administration Console showing the last 90 days of login attempts, session activity, and account security events for a given customer account.

Single Sign-On (SSO)

A login method allowing customers to authenticate using an existing Google, Apple, or Microsoft account instead of creating a separate TechMart password.

TechMart Connect

The unified account, application, and cloud platform used to manage TechMart Home Hub devices, subscriptions, and API access.

TechMart Connect Sync Service

A background Windows service installed alongside the desktop application that keeps device states synchronized even when the main application window is closed.

Webhook

An automated HTTP callback configured by a developer to receive real-time notifications of device events from the TechMart cloud platform.

Webhook Delivery Log

A record in the Developer Console showing the status, response code, and retry history of every webhook notification sent to a developer's configured endpoint.

Appendix H: Sample Escalation Scenarios and Resolutions

Document ID: TM-TECH-ESCALATION-H1 | TechMart Electronics Technical Support Knowledge Base | Category: Technical Support

The following anonymized, composite scenarios illustrate how the diagnostic material in this knowledge base is applied to real support tickets from intake through resolution. These examples are intended as training reference material for new support agents.

Scenario: Customer locked out after moving to a new country

A long-time customer relocated from the United States to Germany and reported being unable to log in from their new location, receiving error AUTH-400. Investigation confirmed the account was still assigned to the Americas cluster. Because cross-cluster migration is not a support-agent action, the ticket was routed to Platform Engineering's Data Residency Migration queue, and the agent set expectations with the customer that the process could take up to two weeks, while confirming the customer could continue using the Americas regional login URL in the meantime without any loss of functionality.

Scenario: Repeated firmware recovery mode after a power outage

A customer's Home Hub entered firmware recovery mode three times within a week, each time following a regional power outage reported by the customer's utility company. Rather than repeatedly walking the customer through the USB-C recovery process, the agent identified the pattern, recommended a small uninterruptible power supply (UPS) for the Hub, and flagged the repeated firmware interruption pattern to the Firmware Engineering team, since Gen 3 Hubs were expected to resume interrupted downloads rather than require manual recovery, indicating a possible firmware regression.

Scenario: Developer account suspended for rate-limit abuse during a product launch

A third-party developer's application was auto-suspended (AUTH_APP_SUSPENDED) after exceeding the rate limit more than 50 times in 24 hours during an unexpected traffic spike from their own product launch. The agent verified the traffic was legitimate rather than malicious by reviewing the request patterns in the Developer Console, reinstated the application, and proactively recommended the developer request a temporary rate-limit increase in advance of any future launches.

Scenario: Family Sharing member permanently sees stale device states

One member of a household consistently saw outdated device states while other family members did not. Standard reconnect steps did not resolve the issue. Deeper investigation in the Security Timeline revealed the affected user's account had an extremely old, cached SSO token that had never been refreshed due to a bug in a specific 3.x version of the Android app that the customer had not updated in over a year; forcing a full sign-out and reinstall resolved the issue immediately.

Appendix I: Standard Canned Response Macros

Document ID: TM-TECH-MACROS-I1 | TechMart Electronics Technical Support Knowledge Base | Category: Technical Support

The following canned response macros are available in the support ticketing system and should be personalized with the customer's name and specific details before sending. They are provided here as reference so agents understand the exact wording customers receive and can build on it consistently across channels.

MACRO: LOGIN_LOCKOUT_EXPLAIN

Thanks for reaching out. I can see your account was temporarily locked after several login attempts, which is a standard security measure. This will clear automatically in 30 minutes, or I can send you an unlock link right now if you'd like to get back in sooner. Just let me know which you'd prefer.

MACRO: PASSWORD_RESET_SENT

I've just sent a password reset link to the email address on file. It will be valid for 20 minutes, so please check your inbox (and spam folder, just in case) and click the link as soon as you can. Let me know if it doesn't arrive within a few minutes.

MACRO: MFA_BACKUP_CODE_ISSUED

Since you're unable to access your authenticator app right now, I've issued you a one-time backup code to get logged back in: this code will only work once, so please use it right away and then set up MFA again on a device you have consistent access to.

MACRO: HUB_RED_LED_TROUBLESHOOT

A solid red light on your Home Hub usually means it couldn't connect to your Wi-Fi network. Could you double-check that you're connecting to your 2.4GHz network (not 5GHz) and that the Wi-Fi password was entered correctly? I'm happy to walk through the setup steps together if that's easier.

MACRO: FIRMWARE_RECOVERY_INSTRUCTIONS

It looks like your Hub may have entered firmware recovery mode, which can happen if a firmware update gets interrupted. Not to worry, this is fully recoverable. You'll just need a USB-C cable and a computer to run our free Recovery Tool. I'll send you the link and walk you through each step.

MACRO: RATE_LIMIT_APOLOGY_DEV

Thanks for flagging this. I can see your application hit our rate limit during a traffic spike, which triggered an automatic, temporary suspension as a protective measure. I've reinstated your application now. For future launches, feel free to reach out in advance and we can look at a temporary limit increase for you.

MACRO: ESCALATE_TO_TIER2

I want to make sure this gets the right level of attention, so I'm escalating your case to our Tier 2 technical team, who have deeper diagnostic tools for this kind of issue. They'll follow up directly within one business day. Thank you for your patience while we get this fully sorted out.

MACRO: WARRANTY_RMA_APPROVED

I've reviewed your account and confirmed your Home Hub is still within its one-year warranty period, so I've gone ahead and approved a replacement. You'll receive a prepaid return label by email shortly, and the new unit typically ships within 3 to 5 business days of us receiving the original device back.

MACRO: IDENTITY_RECOVERY_TIMELINE

I understand how frustrating it is to be locked out, and I want to be upfront with you: because you've lost access to both your email and your authenticator, we do need to go through our full identity verification process to protect your account. This is reviewed by a specialized team and typically takes 3 to 5 business days. I'll get your request submitted right away.

MACRO: REGIONAL_OUTAGE_NOTICE

Thanks for reporting this. We're currently aware of a service disruption affecting your region, and our engineering team is actively working on it. There's nothing further needed on your end right now, this isn't specific to your account or device. I'll keep this ticket open and follow up as soon as it's resolved.

Appendix J: Full Sample Support Ticket Transcripts

Document ID: TM-TECH-TRANSCRIPTS-J1 | TechMart Electronics Technical Support Knowledge Base | Category: Technical Support

The following anonymized transcripts are provided as end-to-end training examples showing how the diagnostic knowledge in this document is applied conversationally, from intake to resolution, across different support channels.

J.1 Chat Transcript: Login Lockout

Customer: I keep trying to log in and it says my account is locked. I only tried like 3 times.

Agent: Thanks for letting me know, I'm sorry for the trouble. I can see your account was locked after a few failed attempts within a short window, which our system does automatically for security. The good news is this clears on its own in 30 minutes, or I can send an unlock link right now if you'd like to get back in sooner.

Customer: Can you just send the link, I need to check something quickly.

Agent: Absolutely, sending that now to the email on file. It'll be valid for 20 minutes, so please use it soon. While that's on its way, just to confirm, were you using the correct region login page? I noticed your account is on our Europe cluster.

Customer: Oh, I was on the .com one, not the .eu one. That's probably why it kept failing before too.

Agent: That would do it. Once you're back in using the eu.techmart-connect.example page, you shouldn't see that again. Let me know once you're in and I'm glad to help with whatever else you needed to check.

J.2 Email Transcript: Home Hub Offline After Router Upgrade

Customer: Hi, I upgraded my router last week and now my Home Hub shows offline in the app. Everything else on my network works fine. The light on the Hub is solid red.

Agent: Thanks for the detailed description, that's really helpful. A solid red light means the Hub lost its Wi-Fi connection, which is common after a router change since the network name or password may be different now, or the new router might be broadcasting a combined 2.4GHz/5GHz network and the Hub only supports 2.4GHz. Could you check your new router's settings to confirm whether 2.4GHz and 5GHz are on separate network names, or combined into one?

Customer: They're combined into one name actually.

Agent: That's very likely the cause. I'd recommend splitting them into two separate network names temporarily, just for the Hub's setup process, connecting the Hub to the 2.4GHz one, and you're welcome to recombine them afterward if you prefer. I'll send you a quick guide on how to update the Wi-Fi network on your existing Hub without a full factory reset.

Customer: Got it, will try that tonight. Thanks!

Agent: You're welcome! Feel free to reply here if it's still showing red afterward and we'll dig further.

J.3 Developer Support Transcript: Webhook Failures

Developer: Our webhook endpoint stopped receiving events about two days ago. No errors in our logs though.

Agent: Thanks for reaching out. I checked your Webhook Delivery Log and I can see deliveries have actually been failing on our end with repeated timeouts, which after 500 consecutive failures triggers an automatic disable of the subscription to prevent unnecessary load, and that's what happened here two days ago.

Developer: That would explain the silence. We did move our endpoint behind a new load balancer around then, could that be it?

Agent: Very possibly, especially if the new load balancer's health check or TLS configuration changed. I'd recommend testing your endpoint directly with our Webhook Inspector tool in the Developer Console to confirm it's responding within our 10-second timeout window with a valid TLS certificate. Once you've confirmed that's working, I can manually re-enable your subscription right away.

Developer: Just tested it, response time looks fine now, must have been a temporary certificate issue during the migration.

Agent: Great, I've re-enabled your subscription now. You should start seeing new events come through immediately, and I'd recommend keeping an eye on the Delivery Log for the next day just to be sure everything's stable.

Appendix K: Service Level Agreements and Escalation Matrix

Document ID: TM-TECH-SLA-K1 | TechMart Electronics Technical Support Knowledge Base | Category: Technical Support

This appendix defines the target response and resolution times for each ticket priority level, along with the internal escalation path for issues that cannot be resolved at the initial support tier. These targets apply to consumer accounts; TechMart for Business customers on an Enterprise plan have separately contracted SLAs documented in their individual service agreements.

K.1 Ticket Priority Definitions

Priority	Definition	First Response Target	Resolution Target
P1 - Critical	Account compromise, data loss, or platform-wide outage	15 minutes	4 hours
P2 - High	Single customer fully unable to access account or all devices	2 hours	1 business day
P3 - Standard	Partial functionality issue, single device or feature affected	24 business hours	3 business days
P4 - Low	General question, minor cosmetic issue, or feature request	1 business day	5 business days

K.2 Escalation Path by Category

Category	Tier 1 Owner	Tier 2 Escalation	Tier 3 / Engineering
Login and Authentication	General Support	Identity and Access Team	Platform Engineering
Password and Account Recovery	General Support	Trust and Safety	Platform Engineering
Installation and Setup	General Support	Hardware Support Team	Firmware Engineering
Connectivity and Troubleshooting	General Support	Network Diagnostics Team	Platform Engineering
Developer API and Webhooks	Developer Support	API Platform Team	Platform Engineering
Billing and Subscriptions	General Support	Billing Operations	Finance Systems

Tickets should never be escalated directly to Tier 3 Engineering without first passing through the appropriate Tier 2 team, except for confirmed P1 incidents, where any agent may page the on-call engineering rotation directly through the internal incident management tool to minimize response time.

K.3 Regional Support Hours

Region	Standard Support Hours	P1 Coverage
Americas	6:00 AM to 10:00 PM Eastern, 7 days a week	24/7 on-call rotation
Europe	7:00 AM to 9:00 PM Central European Time, 7 days a week	24/7 on-call rotation
Asia-Pacific	8:00 AM to 8:00 PM Singapore Time, 7 days a week	24/7 on-call rotation

Outside of standard support hours, customer-facing chat and phone support transition to a reduced overnight team focused on triage and P1/P2 handling, with P3 and P4 tickets queued for the next business day in the customer's regional cluster. Developer Support operates Americas business hours only, Monday through Friday, with P1 developer-impacting incidents (such as a full API outage) still covered by the 24/7 platform on-call rotation regardless of the Developer Support team's own hours.

Appendix L: Customer Personas for Support Context

Document ID: TM-TECH-PERSONAS-L1 | TechMart Electronics Technical Support Knowledge Base | Category: Technical Support

Understanding the range of customers who contact TechMart support helps agents calibrate both the technical depth and the tone of their responses. The following personas, developed from support ticket analysis and customer research, are referenced informally throughout agent training materials.

Persona: 'First-Time Sam' - The New Smart Home Owner

Sam recently purchased their first Home Hub as a gift or a personal upgrade and has little to no prior smart home experience. Sam is likely to describe problems in non-technical terms ('the light thing isn't working') and benefits most from short, numbered steps rather than technical explanations of underlying causes. Agents should avoid jargon like 'MQTT' or 'OAuth' entirely when assisting this persona, and should confirm each step was completed successfully before moving to the next.

Persona: 'Power-User Priya' - The Advanced Home Automation Enthusiast

Priya manages a large household with 30 or more connected devices, multiple Hubs, and complex custom automations. Priya is comfortable with networking terminology, has often already tried basic troubleshooting steps before contacting support, and becomes frustrated by being walked through steps she has already attempted. Agents should ask targeted diagnostic questions early (app version, LED status, router model) rather than starting from the most basic possible explanation.

Persona: 'Developer Dev' - The Third-Party Integration Builder

Dev is building a commercial or hobbyist application against the TechMart Developer API and contacts Developer Support specifically, not General Support. Dev communicates in terms of HTTP status codes, error payloads, and SDK versions, and expects precise, technically accurate answers, including exact error code names rather than paraphrased descriptions. Vague reassurances without a specific technical explanation tend to erode Dev's confidence in the platform.

Persona: 'Business Owner Blake' - The Small Business Administrator

Blake manages a TechMart for Business account for a retail location or small office and is primarily concerned with minimizing downtime and protecting shared organizational data, such as during the employee onboarding and offboarding scenarios described in Sections 2.6 and 2.8. Blake values clear communication about SLAs and escalation timelines, as described in Appendix K, and should be proactively informed of relevant self-service admin tools that reduce Blake's future dependency on support.

Persona: 'Concerned Casey' - The Security-Conscious Customer

Casey contacts support specifically about account security, unfamiliar login activity, or privacy questions, and is often anxious about a possible compromise even when a false alarm. Casey should be reassured with concrete, verifiable actions (reviewing the Security Timeline together, confirming MFA is active) rather than generic reassurance alone, and agents should proactively offer to enable MFA if it is not already active, even if that was not the original reason for contact.

These personas are not mutually exclusive, and a single customer may shift between them depending on the specific issue; a Power-User Priya contacting support about a Developer API integration for a personal project, for example, should still receive the precise, technically detailed treatment appropriate to that specific request.

Appendix M: Document Revision History

Document ID: TM-TECH-REVHIST-M1 | TechMart Electronics Technical Support Knowledge Base | Category: Technical Support

This knowledge base is maintained by the TechMart Support Enablement team and updated on a rolling basis as product features, error codes, and support policies change. The revision history below reflects the major updates leading to the current version.

Version	Date	Summary of Changes
1.0	Initial internal release	First consolidated version combining previously separate login, password, and ins
2.0	Following the 2024 platform migration	Rewrote Section 1 and Section 2 for the new OAuth 2.0 authentication system; ac
2.5	After Home Hub Gen 3 launch	Added Gen 3 hardware specifications, updated LED status table, added firmware
3.0	Developer API v2 release	Full rewrite of Section 5 to reflect typed SDK exceptions, webhook signature verifi
3.1	Minor update	Added Family Sharing troubleshooting, Appendix H escalation scenarios, and App
3.2	Current version	Added extended FAQ appendices, canned response macros, sample ticket transo

Support agents identifying outdated information in this document should flag it through the standard Knowledge Base Correction workflow rather than editing customer-facing content directly, ensuring all changes are reviewed for accuracy before publication across the entire support organization.

Appendix N: Quick Reference Contact and Resource Links

Document ID: TM-TECH-QUICKREF-N1 | TechMart Electronics Technical Support Knowledge Base | Category: Technical Support

The following is a consolidated quick-reference list of internal and customer-facing resources mentioned throughout this document, gathered here for convenience during live support interactions.

Resource	Location / Address	Purpose
Account Administration Console	Internal tool (agent access only)	Account lookup, session management, identity verification
Security Timeline	Within Account Administration Console	90-day login and security event history per account
Internal Status Dashboard	Internal tool (agent access only)	Real-time regional and service health status
Developer Console	developers.techmart-electronics.example	API credentials, usage monitoring, webhook configuration
TechMart Recovery Tool	recovery.techmart-electronics.example	Firmware recovery for Home Hub devices via USB
TechMart for Business Portal	business.techmart-electronics.example	Organization admin console, MSI installer, Pre-Provisioning
Downloads Page	downloads.techmart-electronics.example	Windows and macOS desktop installers
Support Sending Address	no-reply@techmart-electronics.example / support@techmart-electronics.example	Automated notifications and customer replies

Agents should bookmark this page during onboarding and refer back to it in the first few weeks of handling tickets until these resource locations become second nature.